Comparative Study of Lightweight Deep Learning Architectures for Responsible Edge AI Deployment in Semiconductor Defect Classification

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by

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DECLARATION

I, **Prasad Malai** (**Roll No: 2023MCS320002**]), hereby declare that, this report entitled **"Comparative Study of Lightweight Deep Learning Architectures for Responsible Edge AI Deployment in Semiconductor Defect Classification"** submitted to Indian Institute of Information Technology Kottayam towards partial requirement of **Master of Technology** in **Computer Science and Engineering with Specialization in Big Data & Machine Learning** is an original work carried out by me under the supervision of **Dr. Ebin Deni Raj** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold the academic ethics and honesty. Whenever an external information or statement or result is used then, that have been duly acknowledged and cited.

Kottayam-686635 **[Prasad Malai]**

May 2026

CERTIFICATE

This is to certify that the work contained in this project report entitled **"Comparative Study of Lightweight Deep Learning Architectures for Responsible Edge AI Deployment in Semiconductor Defect Classification"** submitted by **Prasad Malai** (**Roll No: 2023MCS320002**) to Indian Institute of Information Technology Kottayam towards partial requirement of **Master of Technology** in **Computer Science and Engineering with Specialization in Big Data & Machine Learning** has been carried out by him under my supervision and that it has not been submitted elsewhere for the award of any degree.

Kottayam-686635 (Dr. Ebin Deni Raj)

May 2026 Project Supervisor

ABSTRACT

Semiconductor defect classification plays a crucial role in ensuring product quality and yield in modern manufacturing processes. This work explores the feasibility of deploying edge AI–based solutions for accurate and efficient defect classification, addressing the challenges of limited computational resources and real-time inference requirements. A comparative study of convolutional neural network architectures, including MobileNetV2, ShuffleNet, and ResNet18, is conducted for multi-class defect classification. The models are trained and subsequently optimized for edge deployment using techniques such as model quantization and ONNX-based conversion, resulting in reduced latency and enhanced cross-platform portability. Model performance is evaluated in terms of classification accuracy, inference efficiency, and suitability for real-time edge deployment. To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to visualize the regions influencing model predictions. Quantitative evaluation of the Grad-CAM outputs is performed using metrics such as Mean Intersection over Union (IoU) and Activation Ratio, enabling assessment of localization quality. Experimental results demonstrate that ResNet18 achieves superior localization performance due to its deeper architecture, while MobileNetV2 and ShuffleNet offer a favorable trade-off between computational efficiency and accuracy. The findings highlight the importance of balancing model complexity and interpretability for responsible edge AI deployment in semiconductor defect classification.

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Chapter 1

Introduction

The rapid advancement of semiconductor technology has led to a growing demand for high-performance and reliable integrated circuits. In modern semiconductor fabrication, defect detection and classification play a critical role in maintaining production efficiency and ensuring product reliability. Despite the availability of advanced inspection systems, the increasing complexity and variability of defect patterns continue to pose significant challenges to conventional approaches. Although deep learning techniques have demonstrated promising results in automated defect analysis, their deployment in real-world environments is often constrained by high computational requirements and limited interoperability across platforms. This has led to increasing interest in lightweight, edge-deployable solutions that can operate efficiently in resource-constrained and heterogeneous environments.

1.1 Background of Defect Classes

1.1.1 Semiconductor Manufacturing Overview

Semiconductor manufacturing is a highly complex, multi-stage process used to fabricate integrated circuits (ICs) on silicon wafers. It involves repeated cycles of deposition, lithography, etching, doping, and polishing are performed with nanometer-scale precision.

Each wafer undergoes hundreds of processing steps in a clean room environment. Even extremely small variations in process conditions can lead to defects that affect circuit functionality.

1.1.2 What are Wafer Defects and Why They Occur

Wafer defects refer to unwanted irregularities or anomalies that occur on the surface or within the semiconductor wafer during fabrication. These defects can arise due to:

- Particle Containment in clean rooms
- Equipment malfunction or calibration errors
- Process variation (temperature, pressure, timing) Material imperfections in silicon wafers

1.1.3 Types of Defects

- **Structural Defects:** These are physical damages to the wafer surface.

  Examples:

  - Scratches
  - Cracks
  - Breaks in the wafer surface

  Characteristics:

  - Often caused by handling or mechanical stress
  - Visible in imaging systems
  - May affect entire circuit functionality

- **Contamination Defects:** These are foreign particles or residues on wafer surface. Examples:

  - Dust particles
  - Chemical residues
  - Metallic contamination

  Characteristics:

  - Usually random in distribution
  - Can be very small but still impactful
  - Often introduced during cleaning or processing steps

- **Pattern-Related Defects:** These defects arise when the intended geometric layout of the integrated circuit is incorrectly formed, leading to deviations from the designed pattern. Examples:

  - Bridging (unwanted connection between circuit lines)

- Open circuits (break in expected pattern)
- Misalignment of patterns
- Missing or extra pattern features

Characteristics:

- Occur during lithography or etching steps
- Directly affect circuit behavior
- Harder to detect visually compared to structural defects

1.1.4 Traditional Inspection Methods

- **Optical Inspection Systems:** Automated machines using high-resolution cameras and laser-based imaging to scan wafers.

  How they work:

  - Capture wafer images
  - Compare with reference patterns or neighboring dies
  - Flag deviations as defects

  Limitations:

  - High cost
  - Sensitive to noise
  - Struggles with complex patterns in advanced nodes
  - Requires post-processing interpretation

- **Manual Classification:** Here human experts visually inspect wafer maps or microscope images to identify defect types.

  Limitations:

  - Time-consuming
  - Subjective (depends on expert experience)
  - Not scalable for high-volume production
  - Inconsistent labeling across engineers

Due to the limitations of traditional inspection methods, there has been a significant shift toward automated defect classification using machine learning and deep learning techniques. These approaches aim to improve accuracy, scalability, and real-time decision-making in semiconductor manufacturing environments.

Definition 1.1.1. Some definition....

Theorem 1.1.2. *Some theorem.....*

Proof. Proof is as follows.... □

Corollary 1.1.3. *A corollary to the theorem is....*

Remark 1.1.4. Some remark.....

You may have to type many equations inside the text. The equation can be typed as below.

$$f(x) = \frac{x^2 - 5x + 2}{e^x - 2} = \frac{y^5 - 3}{e^x - 2} \tag{1.1}$$

This can be referred as (1.1) and so on.....

You may have to type a set of equations. For this you may proceed as given below.

$$\begin{aligned} f(x) &= e^{1+2(x-a)} + \dots \\ &= \log(x+a) + \sin(x+y) + \cdots \end{aligned} \tag{1.2}$$

You may have to cite the articles. You may do so as [4] and so on..... Note that you have already created the 'bib.bib' file and included the entry with the above name. Only then you can cite it as above.

1.2 Section-2 Name

Definition 1.2.1. Some definition....

Remark 1.2.2. Some remark.....

1.2.1 Subsection name

Theorem 1.2.3. *Some theorem.....*

Proof. Proof is as follows.... □

[The figure will be displayed here.]

Figure 1.1: The correlation coefficient as a function of ρ

Chapter 2

Chapter-2 Name

Introductory lines...

2.1 Section-1 Name

Definition 2.1.1. Some definition....

Remark 2.1.2. Some remark.....

Theorem 2.1.3. *Some theorem.....*

Proof. Proof is as follows.... □

2.2 Section-2 Name

Definition 2.2.1. Some definition....

Remark 2.2.2. Some remark.....

2.2.1 Subsection name

Theorem 2.2.3. *Some theorem.....*

Proof. Proof is as follows.... □

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